



Analytic Number Theory Note

Introduction to the Theory of Numbers

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Chapter 1 Introduction

1.1 Basic Definitions

Definition 1.1 (Natural Numbers)

The set of natural numbers is denoted by $\mathbb{N} = \{1, 2, 3, \dots\}$.



Definition 1.2 (Prime Numbers)

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself. And the set of prime numbers is denoted by $\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$.



Definition 1.3 (Prime Counting Function)

We using the notation $\pi(N) = \#\{p \leq N : p \in \mathbb{P}\}$ to denote the number of primes less than or equal to N . Sometimes, we also use $\pi(N) = \sum_{p \leq N} 1$ to denote the same function.



Definition 1.4 (Euler Totient Function)

For any natural number n , the Euler totient function $\phi(n)$ is defined as the number of integers from 1 to n that are coprime to n . In other words,

$$\phi(n) = \#\{k \in \mathbb{N} : 1 \leq k \leq n, \gcd(k, n) = 1\}.$$



1.2 Fundamental Theorem of Arithmetic and Infinitely Many Primes

Theorem 1.1 (Fundamental Theorem of Arithmetic (FTA))

Every natural number greater than 1 can be uniquely factored into prime numbers, up to the order of the factors. In other words, for any $n \in \mathbb{N}$ with $n > 1$, there exist unique primes $p_1 < p_2 < \dots < p_k$ and positive integers $\alpha_1, \alpha_2, \dots, \alpha_k$ such that

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}.$$



Proof First, we show existence. We proceed by induction on n . The base case $n = 2$ is prime itself. Assume true for all integers less than n . If n is prime, we are done. If not, let $n = ab$ with $1 < a, b < n$. By the induction hypothesis, both a and b can be factored into primes. Combining these factorizations gives a prime factorization of n .

Next, we show uniqueness. Suppose n has two distinct prime factorizations:

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k} = q_1^{\beta_1} q_2^{\beta_2} \dots q_m^{\beta_m},$$

where p_i and q_j are primes. Without loss of generality, let p_1 be the smallest prime in the first factorization. Since p_1 divides n , it must divide the right-hand side. By the property of primes, p_1 must equal one of the q_j . Repeating this argument for all primes in both factorizations shows that the multisets of primes must be identical, establishing uniqueness.

Theorem 1.2 (Infinitely Many Primes)

Over 2,300 years ago, Euclid showed that there are infinitely many primes. (i.e. $\lim_{N \rightarrow \infty} \pi(N) = \infty$)



Proof Assume, for the sake of contradiction, that there are only finitely many primes: $p_1, p_2, \dots, p_k$. Consider the number

$$N = p_1 p_2 \dots p_k + 1.$$

By construction, N is not divisible by any of the primes $p_1, p_2, \dots, p_k$, since it leaves a remainder of 1 when divided by any of them. Therefore, either N is prime itself or it has prime factors not in our original list. In either case, this contradicts

our assumption that we had listed all primes. Hence, there must be infinitely many primes.

Proposition 1.1 (Lower Bound for $\pi(n)$)

For any natural number $n \geq 1$, we have the following inequality for $\pi(n)$:

$$\log \log n \leq \pi(n)$$

Exercise 1.1 Prove the Proposition 1.1. (workshop 1)

(Hint: Let p_n be the n -th prime number. Show that $p_n \leq 2^{2^{n-1}}$ for all $n \geq 1$.)

Solution First, we prove by induction that $p_n \leq 2^{2^{n-1}}$ for all $n \geq 1$.

The base case $n = 1$ holds since $p_1 = 2 \leq 2^{2^0} = 2$. Assume the statement is true for some $k \geq 1$, i.e., $p_k \leq 2^{2^{k-1}}$. We need to show it holds for $k + 1$. Consider the number

$$N = p_1 p_2 \cdots p_k + 1.$$

By construction, N is not divisible by any of the primes $p_1, p_2, \dots, p_k$. Therefore, either N is prime itself or it has a prime factor greater than p_k . In either case, we have

$$p_{k+1} \leq N = p_1 p_2 \cdots p_k + 1 \leq (2^{2^0})(2^{2^1}) \cdots (2^{2^{k-1}}) + 1 = 2^{\sum_{i=0}^{k-1} 2^i} + 1 = 2^{2^k - 1} + 1 \leq 2^{2^k}.$$

Thus, by induction, the inequality $p_n \leq 2^{2^{n-1}}$ holds for all $n \geq 1$.

Now, to prove the proposition, let $n \geq 1$ and let $k = \pi(n)$, which is the number of primes less than or equal to n . By the definition of $\pi(n)$, we have $p_k \leq n < p_{k+1}$. Using the inequality we just proved, we get

$$n < p_{k+1} \leq 2^{2^k} \implies \log n \leq 2^k \log 2 \implies \log \log n \leq k \log 2 + \log \log 2 \leq k = \pi(n).$$

Therefore, we conclude that $\log \log n \leq \pi(n)$ for all $n \geq 1$.

Theorem 1.3 (Prime Number Theorem (PNT))

As N approaches infinity, the prime counting function $\pi(N)$ is asymptotically equivalent to $\frac{N}{\log N}$: $\pi(N) \sim \frac{N}{\log N}$ as $N \rightarrow \infty$. In other words,

$$\lim_{N \rightarrow \infty} \frac{\pi(N)}{\frac{N}{\log N}} = 1.$$

The proof of Theorem 1.3 is one of the main goal of this course and will be given in later chapters after introducing more tools from complex analysis.

Lemma 1.1

To prove Theorem 1.3, it equivalent to show that for any positive constants c_1, c_2 with $c_1 < 1 < c_2$, we have

$$c_1 \frac{N}{\log N} \leq \pi(N) \leq c_2 \frac{N}{\log N}$$

Proof Suppose Theorem 1.3 holds. Then for any $\varepsilon > 0$, there exists N_0 such that for all $N > N_0$,

$$\left| \frac{\pi(N)}{\frac{N}{\log N}} - 1 \right| < \varepsilon.$$

Choosing $\varepsilon = \min(1 - c_1, c_2 - 1)$, we have

$$1 - \varepsilon < \frac{\pi(N)}{\frac{N}{\log N}} < 1 + \varepsilon,$$

which implies

$$c_1 \frac{N}{\log N} < \pi(N) < c_2 \frac{N}{\log N}.$$

Conversely, suppose the inequalities hold for all sufficiently large N . Then for any $\varepsilon > 0$, we can choose $c_1 = 1 - \varepsilon$ and $c_2 = 1 + \varepsilon$. Thus, for all sufficiently large N ,

$$(1 - \varepsilon) \frac{N}{\log N} < \pi(N) < (1 + \varepsilon) \frac{N}{\log N},$$

which leads to

$$-\varepsilon < \frac{\pi(N)}{\frac{N}{\log N}} - 1 < \varepsilon.$$

Therefore,

$$\left| \frac{\pi(N)}{\frac{N}{\log N}} - 1 \right| < \varepsilon,$$

proving Theorem 1.3.

Remark A more precise version of Theorem 1.3 states that

$$\pi(x) = \text{Li}(x) + E(x) \quad (1.1)$$

where $\text{Li}(x) = \int_2^x \frac{dt}{\log t} \sim \frac{x}{\log x}$ and the error term $E(x)$ satisfies $E(x) = \mathcal{O}(\sqrt{x} \log x)$.

 **Exercise 1.2** Prove that equation (1.1) implies Theorem 1.3.

Solution From equation (1.1), we have

$$\pi(x) = \text{Li}(x) + E(x).$$

Dividing both sides by $\frac{x}{\log x}$, we get

$$\frac{\pi(x)}{\frac{x}{\log x}} = \frac{\text{Li}(x)}{\frac{x}{\log x}} + \frac{E(x)}{\frac{x}{\log x}}.$$

As $x \rightarrow \infty$, we know that $\text{Li}(x) \sim \frac{x}{\log x}$, so

$$\lim_{x \rightarrow \infty} \frac{\text{Li}(x)}{\frac{x}{\log x}} = 1.$$

Next, we analyze the error term:

$$\left| \frac{E(x)}{\frac{x}{\log x}} \right| = \left| E(x) \cdot \frac{\log x}{x} \right| = \mathcal{O} \left(\sqrt{x} \log x \cdot \frac{\log x}{x} \right) = \mathcal{O} \left(\frac{(\log x)^2}{\sqrt{x}} \right).$$

Since $\frac{(\log x)^2}{\sqrt{x}} \rightarrow 0$ as $x \rightarrow \infty$, we have

$$\lim_{x \rightarrow \infty} \frac{E(x)}{\frac{x}{\log x}} = 0.$$

Combining these results, we find

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\frac{x}{\log x}} = 1 + 0 = 1.$$

Thus, equation (1.1) implies Theorem 1.3.

Definition 1.5 (Riemann Zeta Function (Outline))

The Riemann zeta function $\zeta(z)$ is defined for complex numbers z and for $\Re(z) > 1$ is given by the infinite series

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}.$$

for the other values of z , $\zeta(z)$ is defined by analytic continuation as it has an analytic continuation to the whole complex plane except $z = 1$.

In the following chapters, we will show that

$$\zeta(1-z) = 2^{1-z} \pi^{-z} \cos\left(\frac{\pi z}{2}\right) \Gamma(z) \zeta(z) \quad \text{for all } z \in \mathbb{C}.$$

and thus $\zeta(-2n) = 0$ which are called the trivial zeros of $\zeta(z)$.



Moreover, if we let $\{\rho\}$ be the set of non-trivial zeros of $\zeta(z)$, then we have the following explicit formula for $\pi(x)$:

$$\pi(x) = \text{Li}(x) - \sum_{\rho} \text{Li}(x^{\rho}) + \text{smaller order terms}.$$

Proposition 1.2 (Analytic Continuation of Zeta Function)

The Riemann zeta function $\zeta(z)$ is holomorphic for all z with $\Re(z) > 1$.

 **Exercise 1.3** Show the proposition 1.2 by Weierstrass M-test.

Solution To show that the Riemann zeta function $\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$ is holomorphic for all z with $\Re(z) > 1$, we will use the Weierstrass M-test.

Let $z = x + iy$ where $x = \Re(z)$ and $y = \Im(z)$. For $\Re(z) > 1$, we have $x > 1$. We can write

$$\left| \frac{1}{n^z} \right| = \frac{1}{|n^z|} = \frac{1}{n^{\Re(z)}} = \frac{1}{n^x}.$$

Since $x > 1$, the series $\sum_{n=1}^{\infty} \frac{1}{n^x}$ converges.

Now, we can choose $M_n = \frac{1}{n^x}$ as our bounding sequence. Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^x}$ converges for $x > 1$, by the Weierstrass M-test, the series $\sum_{n=1}^{\infty} \frac{1}{n^z}$ converges uniformly on any compact subset of the half-plane $\Re(z) > 1$.

Since each term $\frac{1}{n^z}$ is holomorphic in z for $\Re(z) > 1$, and the series converges uniformly, it follows that the sum $\zeta(z)$ is also holomorphic in this region.

Therefore, we conclude that the Riemann zeta function $\zeta(z)$ is holomorphic for all z with $\Re(z) > 1$.

Postulate 1.1 (Riemann Hypothesis (RH))

All non-trivial zeros ρ of the Riemann zeta function $\zeta(z)$ have real part equal to $\frac{1}{2}$, i.e., $\Re(\rho) = \frac{1}{2}$.

Remark One can find that the non-trivial zeros of $\zeta(z)$ are only appear in the critical strip: $0 < \Re(z) < 1$.

Definition 1.6

Usually, we using the notation $(RH)_{\sigma}$ to denote the following statement:

$$\zeta(\rho) \neq 0 \quad \text{for all } \rho \text{ with } \Re(\rho) > \sigma.$$

where ρ is the set of non-trivial zeros of $\zeta(z)$.

By extend Euler Product Formula 1.4 to complex variable, one can show that $\zeta(z) \neq 0$ for all z with $\Re(z) > 1$. Hence, $(RH)_1$ is true. And the Riemann Hypothesis 1.1 is equivalent to $(RH)_{\frac{1}{2}}$.

A fact is that for $\sigma < 1$, $(RH)_{\sigma}$ is equivalent to the following statement about the error term $E(x)$ in equation (1.1):

$$|\pi(x) - \text{Li}(x)| = \mathcal{O}(x^{\sigma} \log x).$$

and by using this fact, it can be shown the Prime Number Theorem 1.3. But unfortunately, we have no ideal if $(RH)_{\sigma}$ is true for any $\sigma < 1$.

However, Hadamard and de la Vallée Poussin (1896) independently proved that by using the fact that $(RH)_1$ is true (i.e. $\zeta(z) \neq 0$ for $\Re(z) > 1$), one can prove the Prime Number Theorem 1.3.

 **Note** Why the zeros of $\zeta(z)$ is important for the distribution of primes?

From the Complex Analysis course, we know that a holomorphic function is completely determined by its zeros and poles (i.e. Singularities). In order to using complex analysis to study $\zeta(z)$, one introduces a fixed function which has the same zeros and poles as $\zeta(z)$, called the Riemann Xi function $\xi(z)$:

$$\xi(z) = \frac{1}{2} z(z-1) \pi^{-\frac{z}{2}} \Gamma\left(\frac{z}{2}\right) \zeta(z).$$

where $\Gamma(z)$ is the Gamma function.

A famous result about $\xi(z)$ is the Hadamard Factorization Theorem:

$$\xi(z) = e^{A+Bz} \prod_{\rho} \left(1 - \frac{z}{\rho}\right) e^{\frac{z}{\rho}}$$

where the product is over all non-trivial zeros ρ of $\zeta(z)$, and A and B are constants.

So once we know the distribution of zeros of $\zeta(z)$, we can understand the behavior of $\xi(z)$ and hence $\zeta(z)$, which in turn gives us information about the distribution of prime numbers via the explicit formula for $\pi(x)$.

1.3 Euler Product Formula and Dirichlet Theorem

Theorem 1.4 (Euler Product Formula)

Thinking the zeta function as a function of real variable $x > 1$, we have the Euler Product Formula:

$$\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-x}} \quad \text{for } x > 1.$$



Proof Recall the geometric series formula: for $|r| < 1$, we have

$$\frac{1}{1-r} = \sum_{k=0}^{\infty} r^k = 1 + r + r^2 + r^3 + \dots$$

Setting $r = p^{-x} < 1$ for each prime p , we get

$$\frac{1}{1-p^{-x}} = \sum_{k=0}^{\infty} p^{-kx} = 1 + p^{-x} + p^{-2x} + p^{-3x} + \dots$$

Consider two distinct primes p and q . Multiplying their corresponding series, we have

$$(1 + p^{-x} + p^{-2x} + \dots)(1 + q^{-x} + q^{-2x} + \dots) = \sum_{j,k \geq 0} \frac{1}{[p^j q^k]^x}$$

and so by the Fundamental Theorem of Arithmetic 1.1, the number of the form $p^j q^k$ are precisely the number n whose prime factors are only p and q . Hence

$$\frac{1}{1-p^{-x}} \cdot \frac{1}{1-q^{-x}} = \sum_{n \in N_{p,q}} \frac{1}{n^x} \quad \text{where } N_{p,q} = \{n \in \mathbb{N} : n = p^j q^k, j, k \in \mathbb{N}_0\}.$$

Extending this argument to all primes less than or equal to integer N , let $E_N = \{n \in \mathbb{N} : \text{all prime factors of } n \text{ are } \leq N\}$, we have

$$\prod_{p \leq N} \frac{1}{1-p^{-x}} = \sum_{n \in E_N} \frac{1}{n^x}.$$

Now, we show that as $N \rightarrow \infty$, we can get Euler Product Formula. Let

$$S_N = \sum_{n \in E_N} \frac{1}{n^x} \quad T_N = \zeta(x) - S_N = \sum_{n \notin E_N} \frac{1}{n^x}.$$

For any $n \notin E_N$, n must have a prime factor greater than N . Thus, $n = pm$ for some prime $p > N$ and some $m \in \mathbb{N}$. Therefore,

$$T_N \leq \sum_{\substack{p > N \\ p \in \mathbb{P}}} \sum_{m=1}^{\infty} \frac{1}{(pm)^x} = \sum_{\substack{p > N \\ p \in \mathbb{P}}} \frac{1}{p^x} \sum_{m=1}^{\infty} \frac{1}{m^x} = \zeta(x) \sum_{\substack{p > N \\ p \in \mathbb{P}}} \frac{1}{p^x}.$$

Since the series $\sum_{p \in \mathbb{P}} \frac{1}{p^x}$ converges for $x > 1$ by p-series test and comparison test, the tail $\sum_{p > N} \frac{1}{p^x} \rightarrow 0$ as $N \rightarrow \infty$. Hence, $T_N \rightarrow 0$ as $N \rightarrow \infty$, which implies that $S_N \rightarrow \zeta(x)$. Thus we have

$$\zeta(x) = \lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \prod_{\substack{p \leq N \\ p \in \mathbb{P}}} \frac{1}{1-p^{-x}} = \prod_{p \in \mathbb{P}} \frac{1}{1-p^{-x}}.$$

Corollary 1.1 (Divergence of the Sum of Reciprocals of Primes)

By taking the logarithmic derivative of the Euler Product Formula, we can get the following fact

$$\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty.$$



Proof Taking the logarithm of both sides of the Euler Product Formula, we have

$$\log \zeta(x) = - \sum_{p \in \mathbb{P}} \log(1 - p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{p^{-kx}}{k}$$

where the last equality follows from the Taylor series expansion of $\log(1 - y) = -\sum_{k=1}^{\infty} \frac{y^k}{k}$ for $0 \leq y < 1$.

For the inner sum, we have

$$\sum_{k=2}^{\infty} \frac{p^{-kx}}{k} < \sum_{k=2}^{\infty} p^{-k} = \frac{p^{-2}}{1 - p^{-1}} = \frac{1}{p(p-1)}.$$

Thus,

$$\log \zeta(x) = \sum_{p \in \mathbb{P}} \left(\frac{p^{-x}}{1} + \sum_{k=2}^{\infty} \frac{p^{-kx}}{k} \right) = \sum_{p \in \mathbb{P}} p^{-x} + \sum_{p \in \mathbb{P}} \sum_{k=2}^{\infty} \frac{p^{-kx}}{k} \leq \sum_{p \in \mathbb{P}} p^{-x} + E(x)$$

where $E(x) = \sum_{p \in \mathbb{P}} \frac{1}{p(p-1)}$ is a convergent series, and

$$E(x) \leq \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1.$$

Therefore,

$$\log \zeta(x) \leq \sum_{p \in \mathbb{P}} p^{-x} + 1.$$

As $x \rightarrow 1^+$, $\zeta(x) \rightarrow \infty$, so $\log \zeta(x) \rightarrow \infty$ by continuity of logarithm. Hence,

$$\sum_{p \in \mathbb{P}} p^{-x} \rightarrow \infty \quad \text{as } x \rightarrow 1^+.$$

and thus,

$$\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty.$$

Definition 1.7 (Arithmetic Progression (AP))

An arithmetic progression is a sequence of numbers in which the difference between consecutive terms is constant.

It can be expressed in the form:

$$\text{AP}_{a,q} = \{a + q, a + 2q, a + 3q, \dots\} = \{nq + a : n \in \mathbb{N}\}$$

where a is the first term and q is the common difference.



Proposition 1.3 (Finiteness of Primes in Certain APs)

If $d = \gcd(a, q) > 1$, then the arithmetic progression $\text{AP}_{a,q} = \{nq + a : n \in \mathbb{N}\}$ contains only finitely many primes.



Exercise 1.4 Prove the proposition above.

Solution If $d = \gcd(a, q) > 1$, then every term in the arithmetic progression can be expressed as

$$nq + a = n(dk) + (dm) = d(nk + m)$$

for some integers k and m . This means that every term in the progression is divisible by d . Since $d > 1$, none of these terms can be prime (except possibly the first term if it equals d itself). Therefore, the arithmetic progression contains only finitely many primes.

Theorem 1.5 (Dirichlet's Theorem on Arithmetic Progressions)

If a and q are coprime (i.e., $\gcd(a, q) = 1$), then the arithmetic progression $\text{AP}_{a,q} = \{nq + a : n \in \mathbb{N}\}$ contains infinitely many primes.



The proof of Dirichlet's Theorem requires more advanced tools from analytic number theory, including Dirichlet characters and L-functions, which will be introduced in later chapters. However, various special cases and consequences of this theorem can be explored with more elementary methods. The following exercise provides an opportunity to investigate one such special case.

Exercise 1.5 Give an elementary proof of Dirichlet's Theorem for the AP $\{4n - 1 : n \in \mathbb{N}\}$ by completing the following

outline:

- Argue by contradiction as Euclid did for the infinitude of primes and suppose there are only finitely many primes of the form $4n - 1$, say $p_1, p_2, \dots, p_k$.
- Consider the number $N := 2^2 \cdot 3 \cdot 5 \cdots p_k - 1$. (all odd primes up to p_k).
- Note that $N > p$ and $N = 4m - 1$ for some $m \in \mathbb{N}$.
- Hence N is not prime. Deduce that N has a prime factor of the form $4n - 1$ to obtain a contradiction.

Solution We proceed by contradiction. Suppose there are only finitely many primes of the form $4n - 1$, say $p_1, p_2, \dots, p_k$. Consider the number

$$N := 2^2 \cdot 3 \cdot 5 \cdots p_k - 1.$$

Note that $N > p_k$ since $p_k \geq 3$. Also, we can express N in the form $4m - 1$ for some integer m .

Since N is greater than any of the primes $p_1, p_2, \dots, p_k$, it cannot be prime itself (as it would contradict our assumption). Therefore, N must be composite and have prime factors.

Now, we analyze the prime factors of N . Any prime factor of N must be either of the form $4n - 1$ or $4n + 1$. (This is because all primes greater than 2 are odd, and odd primes can be expressed in one of these two forms.) However, if all prime factors were of the form $4n + 1$, then their product would also be of the form $4n + 1$. This is because the product of two numbers of the form $4n + 1$ is also of that form:

$$(4a + 1)(4b + 1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1.$$

Since we have established that N is of the form $4m - 1$, it cannot be expressed as a product of only primes of the form $4n + 1$. Therefore, at least one prime factor of N must be of the form $4n - 1$. But this contradicts our initial assumption that $p_1, p_2, \dots, p_k$ were the only primes of that form and all of them not dividing N .

Hence, we conclude that there are infinitely many primes of the form $4n - 1$.

Lemma 1.2

Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\text{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \pmod{q}, \\ 0 & \text{otherwise.} \end{cases}$$

Then once we have

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

we can show that the series

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p} = \infty$$

and hence Dirichlet's Theorem 1.5 holds.



Proof Dirichlet was inspired by Euler's proof that the series of reciprocals of primes diverges (Corollary 1.1) to prove his theorem. Let $\mathbb{1}_{a,q}(n)$ be the indicator function for the arithmetic progression $\text{AP}_{a,q}$, defined as

$$\mathbb{1}_{a,q}(n) = \begin{cases} 1 & \text{if } n \equiv a \pmod{q}, \\ 0 & \text{otherwise.} \end{cases}$$

Then once can show that the series

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p} = \infty$$

for any a, q with $\gcd(a, q) = 1$, which implies that there are infinitely many primes in the arithmetic progression $\text{AP}_{a,q}$ and hence Dirichlet's Theorem 1.5 holds. Then one might hope extend Euler's Product Formula 1.4 to

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

On the other hand, since

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n} = \sum_{m=1}^{\infty} \frac{1}{mq+a} \geq \frac{1}{a+q} \sum_{m=1}^{\infty} \frac{1}{m} = \infty.$$

we see that $L(x; \mathbb{1}_{a,q}) := \sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x}$ tends to infinity as $x \rightarrow 1^+$. Thus, do the similar things as in Corollary 1.1, we have

$$\log L(x; \mathbb{1}_{a,q}) = - \sum_{p \in \mathbb{P}} \log(1 - \mathbb{1}_{a,q}(p)p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k}.$$

where the last equality follows from the Taylor series expansion of $\log(1-y) = -\sum_{k=1}^{\infty} \frac{y^k}{k}$ for $0 \leq y < 1$. And as before, since

$$\sum_{k=2}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k} \leq \sum_{k=2}^{\infty} p^{-k} = \frac{p^{-2}}{1-p^{-1}} = \frac{1}{p(p-1)},$$

we have

$$\log L(x; \mathbb{1}_{a,q}) = \sum_{p \in \mathbb{P}} \left(\frac{\mathbb{1}_{a,q}(p)p^{-x}}{1} + \sum_{k=2}^{\infty} \frac{\mathbb{1}_{a,q}(p)^k p^{-kx}}{k} \right) = \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} + E(x) \quad (1.2)$$

where $E(x) = \sum_{p \in \mathbb{P}} \frac{1}{p(p-1)} \leq 1$ is a convergent series as before. Therefore,

$$\log L(x; \mathbb{1}_{a,q}) \leq \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} + 1.$$

As $x \rightarrow 1^+$, $L(x; \mathbb{1}_{a,q}) \rightarrow \infty$, so $\log L(x; \mathbb{1}_{a,q}) \rightarrow \infty$ by continuity of logarithm. Hence,

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} \rightarrow \infty \quad \text{as } x \rightarrow 1^+.$$

and thus,

$$\sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p} = \infty.$$



Note Note that if we want to show

$$\sum_{n=1}^{\infty} \frac{\mathbb{1}_{a,q}(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \mathbb{1}_{a,q}(p)p^{-x}} \quad \text{for } x > 1.$$

in the above Lemma 1.2. Examining the proof of Euler Product Formula 1.4, we need to show that $\mathbb{1}_{a,q}(n)$ is completely multiplicative. (i.e. $\mathbb{1}_{a,q}(p^j q^k) = \mathbb{1}_{a,q}(p)^j \mathbb{1}_{a,q}(q)^k$). But unfortunately, $\mathbb{1}_{a,q}(n)$ is not completely multiplicative on arithmetic progression. So Lemma 1.2 just gives the idea of Dirichlet's proof of Theorem 1.5, but we need more advanced tools to complete the proof.

One way to fix this problem is to introduce Dirichlet characters χ modulo q , which will be discussed in later chapters. The core idea is to show the formula

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^x} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-x}} \quad \text{for } x > 1.$$

and then using the similar argument as in Lemma 1.2 but replacing $\mathbb{1}_{a,q}(n)$ by Dirichlet characters χ modulo q : $\mathbb{1}_{a,q}(n) = \sum_{\chi} c_{\chi} \chi(n)$ (which will be discussed in later chapters) to get

$$\log L(x; \chi) = - \sum_{p \in \mathbb{P}} \log(1 - \chi(p)p^{-x}) = \sum_{p \in \mathbb{P}} \sum_{k=1}^{\infty} \frac{\chi(p)^k p^{-kx}}{k}.$$

as before, where $L(x; \chi) := \sum_{n=1}^{\infty} \frac{\chi(n)}{n^x}$ and

$$\sum_{\chi} c_{\chi} \log L(x; \chi) = \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{q}}} \frac{1}{p^x} + E(x)$$

as in equation (1.2). At last, we just need $\sum_{\chi} c_{\chi} \log L(x; \chi) \rightarrow \infty$ as $x \rightarrow 1^+$, the following Lemma 1.3 give us $\log L(x; \chi_0) \rightarrow \infty$, and later chapters will show that:

- $L(x; \chi)$ is bounded for all $\chi \neq \chi_0$ as $x \rightarrow 1^+$.
- $L(x; \chi) \neq 0$ for all $\chi \neq \chi_0$ as $x \rightarrow 1^+$.

by using these two properties, we can show that $\sum_{\chi} c_{\chi} \log L(x; \chi) \rightarrow \infty$ as $x \rightarrow 1^+$. Finally, we arrive at the conclusion of Dirichlet's Theorem 1.5.

Lemma 1.3

For principal Dirichlet character χ_0 modulo q

$$\chi_0(n) = \begin{cases} 1 & \text{if } \gcd(n, q) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

we have

$$L(x; \chi_0) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^x} \rightarrow \infty \quad \text{as } x \rightarrow 1^+.$$



Proof By the definition of principal Dirichlet character χ_0 modulo q , we have

$$L(x; \chi_0) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^x} = \sum_{m=0}^{\infty} \sum_{\substack{b=1 \\ \gcd(b, q)=1}}^q \frac{1}{(mq+b)^x} \geq \sum_{m=0}^{\infty} \sum_{\substack{b=1 \\ \gcd(b, q)=1}}^q \frac{1}{(mq+q)^x} = \phi(q) \sum_{m=1}^{\infty} \frac{1}{(mq)^x} = \frac{\phi(q)}{q^x} \sum_{m=1}^{\infty} \frac{1}{m^x}.$$

where the second equality follows from the definition of $\chi_0(n)$, and $\gcd(b, q) = 1$ if and only if $\gcd(mq+b, q) = 1$ for any $m \in \mathbb{N}_0$.

Thus, as $x \rightarrow 1^+$, we have

$$L(x; \chi_0) \geq \frac{\phi(q)}{q^x} \sum_{m=1}^{\infty} \frac{1}{m^x} \rightarrow \infty.$$

 **Exercise 1.6** The Principal Dirichlet Character χ_0 modulo q is defined as

$$\chi_0(n) = \begin{cases} 1 & \text{if } \gcd(n, q) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

Show that

$$\chi_0(mn) = \chi_0(m)\chi_0(n) \quad \text{for all } m, n \in \mathbb{N}.$$

and

$$\chi_0(b+nq) = \chi_0(b) \quad \text{for all } b, n \in \mathbb{N}.$$

Solution First, we show that $\chi_0(mn) = \chi_0(m)\chi_0(n)$ for all $m, n \in \mathbb{N}$. We consider two cases:

(C1) If $\gcd(m, q) = 1$ and $\gcd(n, q) = 1$, then $\gcd(mn, q) = 1$. Therefore,

$$\chi_0(mn) = 1 = 1 \cdot 1 = \chi_0(m)\chi_0(n).$$

(C2) If either $\gcd(m, q) > 1$ or $\gcd(n, q) > 1$, then $\gcd(mn, q) > 1$. Therefore,

$$\chi_0(mn) = 0 = 0 \cdot \chi_0(n) = \chi_0(m)\chi_0(n) \quad \text{or} \quad \chi_0(mn) = 0 = \chi_0(m) \cdot 0 = \chi_0(m)\chi_0(n).$$

Thus, in both cases, we have $\chi_0(mn) = \chi_0(m)\chi_0(n)$.

Next, we show that $\chi_0(b+nq) = \chi_0(b)$ for all $b, n \in \mathbb{N}$. We consider two cases:

(C1) If $\gcd(b, q) = 1$, then $\gcd(b+nq, q) = 1$. Therefore,

$$\chi_0(b+nq) = 1 = \chi_0(b).$$

(C2) If $\gcd(b, q) > 1$, then $\gcd(b+nq, q) > 1$. Therefore,

$$\chi_0(b+nq) = 0 = \chi_0(b).$$

Thus, in both cases, we have $\chi_0(b+nq) = \chi_0(b)$.

Definition 1.8 (Dirichlet L-Function (Outline))

Given a Dirichlet character χ modulo q , the Dirichlet L-function $L(s, \chi)$ is defined for complex numbers s with $\Re(s) > 1$ by the series

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Similar to the Riemann zeta function, the Dirichlet L-function can be analytically continued to the entire complex plane except for a simple pole at $s = 1$ when χ is the principal character. 

1.4 Chebyshev's bounds

Lemma 1.4 (Legendre's Formula)

For any integer n and prime p , consider the prime factorization of $m!$:

$$m! = p^\alpha \cdot (\text{other prime factors not equal to } p).$$

Then we have

$$\alpha = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor.$$

and denote it as $\alpha(p) = \alpha$ of the prime p in the prime factorization of $m!$. 



Note The series $\sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor$ is actually finite since $\left\lfloor \frac{m}{p^k} \right\rfloor = 0$ for all $k > \log_p m$.

Proof Notice that the prime factors of $m!$ are exactly the all prime factors from 1 to m . Since

$$\left\lfloor \frac{m}{p^k} \right\rfloor = \#\{np^k : n \in \mathbb{N}, np^k \leq m\}$$

counts the number of multiples of p^k that are less than or equal to m , each multiple contributes at least k to the exponent α of p in the prime factorization of $m!$. Therefore, summing over all k gives the total exponent of p in $m!$:

$$\alpha = \sum_{k=1}^{\infty} \left\lfloor \frac{m}{p^k} \right\rfloor.$$



Exercise 1.7 Show that $2^n \leq \binom{2n}{n} < 4^n$ for all $n \in \mathbb{N}$.

Solution We can prove this inequality using the definition of the binomial coefficient:

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2}.$$

We want to show that

$$2^n \leq \frac{(2n)!}{(n!)^2}.$$

Multiplying both sides by $(n!)^2$, we need to prove that

$$2^n (n!)^2 \leq (2n)!.$$

We can rewrite $(2n)!$ as the product of the first $2n$ integers:

$$(2n)! = 1 \cdot 2 \cdot 3 \cdots (2n).$$

Now, we can group the terms in pairs:

$$(2n)! = (1 \cdot 2) \cdot (3 \cdot 4) \cdots ((2n-1) \cdot (2n)).$$

Each pair $(2k-1) \cdot (2k)$ for $k = 1, 2, \dots, n$ is greater than or equal to $2k$, since

$$(2k-1) \cdot (2k) = 4k^2 - 2k = 2k(2k-1) \geq 2k \quad \text{for } k \geq 1.$$

Therefore,

$$(2n)! \geq (2 \cdot 1) \cdot (2 \cdot 2) \cdots (2 \cdot n) = 2^n (1 \cdot 2 \cdots n) = 2^n n!.$$

This shows that

$$2^n(n!)^2 \leq (2n)!.$$

For the upper bound, by using the binomial theorem, we have

$$4^n = (1+1)^{2n} = \sum_{k=0}^{2n} \binom{2n}{k} \geq \binom{2n}{n}.$$

Thus, we have shown that

$$2^n \leq \binom{2n}{n} < 4^n.$$

Lemma 1.5 (Chebyshev's Product Bounds lemma)

For n sufficiently large, we have

$$\left(\frac{4}{3}\right)^n \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} \quad \text{and} \quad \prod_{\substack{p \in \mathbb{P} \\ n < p \leq 2n}} p \leq 4^n.$$



Proof By Lemma 1.4, and notice that the prime factors of $m!$ are exactly the all prime factors from 1 to m , we have

$$n! = \prod_{\substack{p \in \mathbb{P} \\ p \leq n}} p^{\alpha(p)} \quad \text{where } \alpha(p) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Thus, by the definition of binomial coefficient, we have

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \frac{\prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\sum_{k=1}^{\infty} \left\lfloor \frac{2n}{p^k} \right\rfloor}}{\left(\prod_{\substack{p \in \mathbb{P} \\ p \leq n}} p^{\sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor} \right)^2} = \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\beta(p)}$$

where

$$\beta(p) = \sum_{k=1}^{\infty} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right).$$

By the **Note**, $\left\lfloor \frac{2n}{p^k} \right\rfloor = 0$ if $p^k > 2n$ or equivalently $k > \log_p(2n)$, and since $0 \leq \lfloor 2x \rfloor - 2\lfloor x \rfloor \leq 1$ for all $x \geq 0$, we have

$$\beta(p) \leq \sum_{k=1}^{\lfloor \log_p(2n) \rfloor} 1 = \lfloor \log_p(2n) \rfloor \leq \log_p(2n) \quad \text{and thus} \quad p^{\beta(p)} = p^{\log_p(2n)} \leq 2n.$$

Furthermore, when $p > \sqrt{2n}$, $\left\lfloor \frac{2n}{p^k} \right\rfloor = 0$ and $\left\lfloor \frac{n}{p^k} \right\rfloor = 0$ for all $k \geq 2$, so

$$\beta(p) = \left\lfloor \frac{2n}{p} \right\rfloor - 2 \left\lfloor \frac{n}{p} \right\rfloor \leq 1 \quad \text{for } p > \sqrt{2n}.$$

Hence,

$$\begin{aligned} \binom{2n}{n} &= \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\beta(p)} = \prod_{\substack{p \in \mathbb{P} \\ p \leq \sqrt{2n}}} p^{\beta(p)} \cdot \prod_{\substack{p \in \mathbb{P} \\ \sqrt{2n} < p \leq 2n}} p^{\beta(p)} \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq \sqrt{2n}}} 2n \cdot \prod_{\substack{p \in \mathbb{P} \\ \sqrt{2n} < p \leq 2n}} p \\ &\leq (2n)^{\sqrt{2n}} \cdot \prod_{\substack{p \in \mathbb{P} \\ \sqrt{2n} < p \leq 2n}} p \\ &\leq \left(\frac{3}{2}\right)^n \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p \end{aligned}$$

where the last inequality holds for n sufficiently large since

$$\lim_{n \rightarrow \infty} \frac{(2n)^{\sqrt{2n}}}{(3/2)^n} = \lim_{n \rightarrow \infty} \frac{e^{\sqrt{2n} \log(2n)}}{e^{n \log(3/2)}} = \lim_{n \rightarrow \infty} e^{n \left(\frac{\log(2n)}{\sqrt{2n}} - \log(3/2) \right)} = 0 \quad \text{since} \quad \lim_{n \rightarrow \infty} \frac{\log(2n)}{\sqrt{2n}} = \lim_{n \rightarrow \infty} \frac{1/n}{1/\sqrt{2n}} = 0.$$

where the last equality follows from L'Hôpital's rule if we treat n as a real variable and then replace it back to integer variable. Therefore, by Exercise 1.7, we have

$$2^n \leq \binom{2n}{n} \leq \left(\frac{3}{2}\right)^n \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p \implies \left(\frac{4}{3}\right)^n \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p.$$

For the second inequality, notice that

$$\beta(p) = \left\lfloor \frac{2n}{p} \right\rfloor - 2 \left\lfloor \frac{n}{p} \right\rfloor = 1 - 0 = 1 \quad \text{for } n < p \leq 2n,$$

Combining with Exercise 1.7 again, we have

$$4^n \geq \binom{2n}{n} = \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p^{\beta(p)} \geq \prod_{\substack{p \in \mathbb{P} \\ n < p \leq 2n}} p.$$

Theorem 1.6 (Chebyshev's Bounds on $\pi(N)$)

There exist positive constants $c_1, c_2 > 0$ such that

$$c_1 \frac{N}{\log N} \leq \pi(N) \leq c_2 \frac{N}{\log N} \quad \text{for } N \text{ sufficiently large.}$$



Proof Chebyshev used the bounds in Lemma 1.5 to prove the theorem. By using the first inequality in Lemma 1.5, we have

$$\left(\frac{4}{3}\right)^n \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} p \leq \prod_{\substack{p \in \mathbb{P} \\ p \leq 2n}} 2n = (2n)^{\pi(2n)}.$$

Taking logarithm on both sides, we have

$$n \log \left(\frac{4}{3}\right) \leq \pi(2n) \log(2n) \implies \frac{\log(4/3)}{2} \cdot \frac{2n}{\log(2n)} \leq \pi(2n).$$

Thus, for $N = 2n$ sufficiently large, it gives the lower bound. And for $N = 2n + 1$ sufficiently large, we have

$$\pi(N) = \pi(2n + 1) \geq \pi(2n) \geq \frac{\log(4/3)}{2} \cdot \frac{2n}{\log(2n)} \geq \frac{\log(4/3)}{4} \cdot \frac{2n + 1}{\log(2n + 1)}$$

where the last inequality holds as $\frac{x}{\log x}$ is increasing for $x \geq e$ ($(\frac{x}{\log x})' = \frac{\log x - 1}{(\log x)^2} > 0$ for $x > e$). Hence, we have the lower bound for all sufficiently large N with $c_1 = \frac{\log(4/3)}{4}$.

For the upper bound, by extend the second inequality in Lemma 1.5 (Let $n = \lfloor \frac{1}{2}x \rfloor, x > 0$), we have

$$\prod_{\substack{p \in \mathbb{P} \\ \lfloor x/2 \rfloor < p \leq 2\lfloor x/2 \rfloor}} p \leq 4^{\lfloor x/2 \rfloor} \leq 4^{x/2} = 2^x$$

and since there are at most one prime between $2\lfloor x/2 \rfloor$ and x . For x sufficiently large, we have

$$\left(\frac{x}{2}\right)^{\pi(x) - \pi(x/2)} \leq \left(\frac{x}{2}\right)^{\pi(x) - \pi(\lfloor x/2 \rfloor)} \leq \prod_{\substack{p \in \mathbb{P} \\ \lfloor x/2 \rfloor < p \leq x}} \frac{x}{2} \leq \prod_{\substack{p \in \mathbb{P} \\ \lfloor x/2 \rfloor < p \leq x}} p \leq x \cdot 2^x \leq 3^x.$$

Taking logarithm on both sides, we have

$$(\pi(x) - \pi(x/2)) \log \left(\frac{x}{2}\right) \leq x \log 3 \implies \pi(x) - \pi(x/2) \leq \frac{x \log 3}{\log(x/2)}.$$

Replacing x by $\frac{x}{2^{j-1}}$ for $j = 1, 2, \dots, m$

$$\pi\left(\frac{x}{2^{j-1}}\right) - \pi\left(\frac{x}{2^j}\right) \leq \frac{\frac{x}{2^{j-1}} \log 3}{\log\left(\frac{x}{2^{j-1}}\right)} \leq \log(9) \cdot \frac{x 2^{-j}}{\log(x 2^{-j})}.$$

Summing over $j = 1, 2, \dots, m$ gives

$$\pi(x) \leq \pi\left(\frac{x}{2^m}\right) + \log(9) \sum_{j=1}^m \frac{x 2^{-j}}{\log(x 2^{-j})}.$$

As for $j \leq m$, we have $\log(x 2^{-j}) \geq \log(x 2^{-m}) \geq \log(x)/2$, $\pi(\frac{x}{2^m}) \leq \frac{x}{2^m} \leq 2\sqrt{x}$ and $\sqrt{x} \leq \frac{x}{\log(x)}$ for x sufficiently

large, it gives

$$\pi(x) \leq 2\sqrt{x} + \log(9) \sum_{j=1}^m \frac{x2^{-j}}{\log(x)/2} = 2\sqrt{x} + \frac{2x \log(9)}{\log(x)} \sum_{j=1}^m 2^{-j} \leq 2\sqrt{x} + \frac{2x \log(9)}{\log(x)} \leq (1 + 2\log(9)) \frac{x}{\log(x)}$$

By setting $c_2 = 1 + 2\log(9)$, and $x = N$, we have the upper bound for all sufficiently large N .

Exercises

- For each $n \in \mathbb{N}$, show that the prime factors of $n(n+1)$ have more than the prime factors of n itself. And deduce that there are infinitely many primes by considering the sequence $\{x_n\}_{n=1}^{\infty}$ where $x_n = x_{n-1}(x_{n-1}+1)$. (Hint: show that $\gcd(n, n+1) = 1$.)

Solution By Euclid's algorithm, we have $\gcd(n, n+1) = \gcd(n+1-n, n) = \gcd(1, n) = 1$.

Thus, n and $n+1$ are coprime, which means they share no common prime factors. Therefore, the prime factors of $n(n+1)$ are the union of the prime factors of n and the prime factors of $n+1$, which means

$$\# \text{prime factors of } n(n+1) = \# \text{prime factors of } n + \# \text{prime factors of } (n+1).$$

and clearly, $\# \text{prime factors of } (n+1) \geq 1$ as it is either prime itself or has at least one prime factor. Hence, the prime factors of $n(n+1)$ have more than the prime factors of n itself.

For the sequence $\{x_n\}_{n=1}^{\infty}$ defined by $x_n = x_{n-1}(x_{n-1}+1)$ with $x_1 = 1$. By previous result, x_n has more (strictly) prime factors than x_{n-1} . Since $x_1 = 1$ has no prime factors, it follows that x_n has at least $n-1$ distinct prime factors. Therefore, as n increases, the number of distinct prime factors of x_n also increases without bound, which implies that there are infinitely many primes.

- Fix any positive integer N , let $P_k = 2, 3, 5, \dots, p_k$ denote the set of first k primes. Define

$$E_N^k = \{n \in \mathbb{N} : n \leq N \text{ and } n \text{ only divisible by any prime in } P_k\}.$$

Show that

- Every $n \in E_N^k$ can be expressed as

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} s^2$$

for some non-negative integers $e_1, e_2, \dots, e_k \in \{0, 1\}$ and some $s \in \mathbb{N}$.

- Then deduce that there are at most $2^k \sqrt{N}$ elements in E_N^k .
- Finally, for any fixed N , let $n \in \mathbb{N}$ such that $P_n = \{p_1, p_2, \dots, p_n\}$ will exact all the primes less than or equal to N . i.e.

$$P_n = \{p \in \mathbb{P} : p \leq N\} = \{2, 3, 5, \dots, p_n\}.$$

So that $\pi(N) = n$ and note that $E_N^n = \{1, 2, \dots, N\}$ (prove it!). From above results, we have $N = \#E_N^n \leq 2^n \sqrt{N}$ or equivalently

$$\pi(N) = n \geq \frac{\log N}{2 \log 2}.$$

By using this, we can show that $\pi(N) \rightarrow \infty$ as $N \rightarrow \infty$. And compare this bound with our previous bound for $\pi(N)$ in Proposition 1.1, which is

$$\pi(N) \geq \log \log N$$

Solution

- By the Fundamental Theorem of Arithmetic 1.1, every $n \in E_N^k$ can be expressed as

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

for some non-negative integers $a_1, a_2, \dots, a_k$. For each a_i , by using Euclid's division algorithm, we can write

$$a_i = 2d_i + r_i \quad \text{where } d_i \in \mathbb{N}_0 \text{ and } r_i \in \{0, 1\}.$$

Thus,

$$n = p_1^{2d_1+r_1} p_2^{2d_2+r_2} \cdots p_k^{2d_k+r_k} = (p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k})^2 \cdot p_1^{r_1} p_2^{r_2} \cdots p_k^{r_k}.$$

and we get the desired form by letting $s = p_1^{d_1} p_2^{d_2} \cdots p_k^{d_k}$.

For the uniqueness of this representation, suppose there are two representations of n :

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} s^2 = p_1^{f_1} p_2^{f_2} \cdots p_k^{f_k} t^2$$

where $e_i, f_i \in \{0, 1\}$ and $s, t \in \mathbb{N}$. Rearranging gives

$$p_1^{e_1-f_1} p_2^{e_2-f_2} \cdots p_k^{e_k-f_k} = \left(\frac{t}{s}\right)^2.$$

The left side is an integer, so $\frac{t}{s}$ must also be an integer, say m . Thus,

$$p_1^{e_1-f_1} p_2^{e_2-f_2} \cdots p_k^{e_k-f_k} = m^2.$$

Since the right side is a perfect square, all exponents on the left side must be even. Given that $e_i, f_i \in \{0, 1\}$, the only way for $e_i - f_i$ to be even is if $e_i = f_i$ for all i . This implies that the exponents are unique.

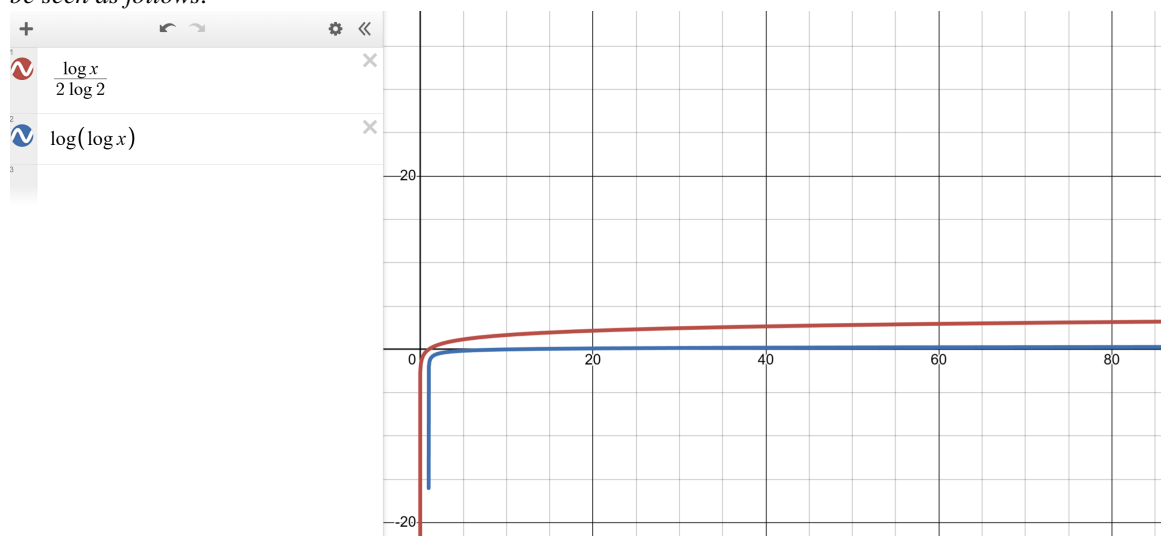
- (b) By the above result, for each $n \in E_N^k$, there are 2^k choices for the exponents $e_1, e_2, \dots, e_k$ and since $n \leq N$, we have

$$s^2 \leq n \leq N \implies s \leq \sqrt{N}.$$

Thus, there are at most $\sqrt{N}$ choices for s . Therefore, the total number of elements in E_N^k is at most $2^k \sqrt{N}$.

- (c) To show that $E_N^n = \{1, 2, \dots, N\}$, notice that any integer m such that $1 \leq m \leq N$ can only have prime factors less than or equal to N . Since P_n contains all primes less than or equal to N , it follows that every integer m in this range is included in E_N^n . Conversely, any number in E_N^n is by definition less than or equal to N . Thus, we have $E_N^n = \{1, 2, \dots, N\}$.

For the lower bound on $\pi(N)$, this bounds shows a quicker growth rate compared to the previous bound. Can be seen as follows:



3. This question show how previous exercises implies that $\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty$. Suppose by contradiction that $\sum_{p \in \mathbb{P}} \frac{1}{p}$ converges. Then the tails of the series must tend to zero, and so we can find a positive integer k such that

$$\sum_{i=k+1}^{\infty} \frac{1}{p_i} < \frac{1}{2}.$$

where we denote the i -th prime by p_i .

Now, fix a positive integer N and fix a prime p then we have

$$\#\{0 < n \leq N : p \mid n, n \in \mathbb{N}\} \leq \frac{N}{p}. \quad (\text{show it!})$$

Then consider E_N^k defined in the previous exercise, we have

$$N - \#E_N^k \leq \frac{N}{p_{k+1}} + \frac{N}{p_{k+2}} + \frac{N}{p_{k+3}} + \cdots < \frac{N}{2}. \quad (\text{show it!})$$

As $\#E_N^k \leq 2^k \sqrt{N}$ from previous exercise, we have

$$\frac{N}{2} < 2^k \sqrt{N} \implies N < 2^{2k+2}$$

which is impossible as N can be arbitrarily large. Hence, we conclude that $\sum_{p \in \mathbb{P}} \frac{1}{p} = \infty$.

Solution

- (a) For a fixed prime p and positive integer N , the integers n such that $1 \leq n \leq N$ and $p \mid n$ are precisely the multiples of p : $p, 2p, 3p, \dots, mp$ where $mp \leq N$. Thus, the largest integer m satisfying this condition is $\lfloor \frac{N}{p} \rfloor$. Therefore we have the inequality.
- (b) For a integer $0 \leq n \leq N$, if $n \notin E_N^k$, then n must be divisible by at least one prime p_i where $i > k$. Thus, by the previous result, the number of integers n such that $1 \leq n \leq N$ and $n \notin E_N^k$ is at most $\frac{N}{p_{k+1}} + \frac{N}{p_{k+2}} + \frac{N}{p_{k+3}} + \dots$. By our assumption, this sum is less than $\frac{N}{2}$. Therefore, we have the inequality.
4. This is a topological way to show there are infinitely many primes. There are some basic concepts in topology we need to introduce first.

Suppose that a collection $\mathcal{B}$ of subsets of a set X (i.e. $\mathcal{B} \subseteq \mathcal{P}(X)$) satisfies the following two properties:

- (i) Every $x \in X$, there is at least one $B \in \mathcal{B}$ such that $x \in B$;
- (ii) For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there is a $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

then we call $\mathcal{B}$ a basis for a topology τ on X it generates. Here a open set of τ is either the empty set or it is a union of sets in $\mathcal{B}$ and a set $F \subset X$ is called closed if its complement $X \setminus F$ is open. We now consider the a topology on $X = \mathbb{Z}$.

- (a) Let $\mathcal{B} = \{S_{a,q} : a, q \in \mathbb{Z}, q > 0\}$ where $S_{a,q} = \{a + nq : n \in \mathbb{Z}\}$ denotes the arithmetic progression. Show that $\mathcal{B}$ is a basis for a topology τ on $\mathbb{Z}$ and any finite set $S \subset \mathbb{Z}$ cannot be open in this topology.
- (b) Show that $S_{a,q}$ is both open and closed for any $a, q \in \mathbb{Z}, q > 0$.
- (c) Finally, show that

$$\mathbb{Z} \setminus \{-1, 1\} = \bigcup_{p \in \mathbb{P}} S_{0,p}.$$

and thus if $\#\mathbb{P} < \infty$, then the right hand side is a closed set but the left hand side cannot be closed. Hence, we conclude that there are infinitely many primes.

Solution

- (a) For any $x \in \mathbb{Z}$, by letting $a = x$ and $q = 1$, then $x \in S_{a,q}$. and hence property (i) holds. For any $B_1 = S_{a_1, q_1}, B_2 = S_{a_2, q_2} \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, then $x = a_1 + n_1 q_1 = a_2 + n_2 q_2$ and by letting $q_3 = \text{lcm}(q_1, q_2)$ and $a_3 = x$. Then $x \in S_{a_3, q_3}$ and for any $y \in S_{a_3, q_3}$, we have

$$y = a_3 + n_3 q_3 = x + n_3 \text{lcm}(q_1, q_2) = x + n_3 m_1 q_1 = x + n'_3 m_2 q_2$$

for some integers m_1, m_2, n_3, n'_3 . Thus, $y \in B_1 \cap B_2$ and hence property (ii) holds. Therefore, $\mathcal{B}$ is a basis for a topology τ on $\mathbb{Z}$.

For any finite set $S \subset \mathbb{Z}$. Suppose by contradiction that S is open, then S is either the empty set (which is not the case here) or it is a union of sets in $\mathcal{B}$. Since each set in $\mathcal{B}$ is infinite, any union of sets in $\mathcal{B}$ must also be infinite. This contradicts the assumption that S is finite. Hence, any finite set $S \subset \mathbb{Z}$ cannot be open in this topology.

- (b) For any $S_{a,q} \in \mathcal{B}$, it is open as it is a union of itself. To show that it is closed, consider its complement

$$\mathbb{Z} \setminus S_{a,q} = \bigcup_{r=0}^{q-1} S_{r,q} \setminus S_{a,q} = \bigcup_{\substack{r=0 \\ r \neq a \pmod q}}^{q-1} S_{r,q}$$

which is a union of sets in $\mathcal{B}$ and hence is open. Therefore, $S_{a,q}$ is closed.

- (c) for each $n \in \mathbb{Z} \setminus \{-1, 1\}$, if $n = 0$, then $n \in S_{0,2}$; if $|n| > 1$, then by the Fundamental Theorem of Arithmetic 1.1, n has at least one prime factor p , so that $n \in S_{0,p}$. Thus $\mathbb{Z} \setminus \{-1, 1\} \subseteq \bigcup_{p \in \mathbb{P}} S_{0,p}$. Conversely, for any $n \in \bigcup_{p \in \mathbb{P}} S_{0,p}$, then there exists a prime p such that $n \in S_{0,p}$, which means $p \mid n$. Therefore, n cannot be ± 1 . Thus, $\bigcup_{p \in \mathbb{P}} S_{0,p} \subseteq \mathbb{Z} \setminus \{-1, 1\}$. Therefore, we get the desired equality.

Chapter 2 Infinite Products and Series

2.1 Infinite Products, Uniform Convergence Criteria and M-test

Definition 2.1 (Infinite Product)

A infinite product $\prod_{n=1}^{\infty} a_n$ converges if the sequence of partial products

$$P_N = \prod_{n=1}^N a_n$$

converges to a limit P as $N \rightarrow \infty$. In this case, we write

$$\prod_{n=1}^{\infty} a_n = P.$$



Theorem 2.1 (n-term Test, Divergence Test for Infinite Products)

If the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit, then $\lim_{n \rightarrow \infty} a_n = 1$.

And thus we usually write infinite products in the form

$$\prod_{n=1}^{\infty} (1 + b_n)$$

and then the n-term test becomes: if $\prod_{n=1}^{\infty} (1 + b_n)$ converges, then $\lim_{n \rightarrow \infty} b_n = 0$.



Proof Suppose that the infinite product $\prod_{n=1}^{\infty} a_n$ converges to a nonzero limit P . Then the sequence of partial products

$$P_N = \prod_{n=1}^N a_n \rightarrow P \neq 0 \quad \text{as } N \rightarrow \infty.$$

Then, $a_n \neq 0$ for any $n \in \mathbb{N}$, and we have

$$a_n = \frac{P_n}{P_{n-1}} \rightarrow \frac{P}{P} = 1 \quad \text{as } n \rightarrow \infty.$$

Definition 2.2 (Bounded Complex-Value Function)

A complex-value function $G : X \rightarrow \mathbb{C}$ defined on a set X is bounded if

$$\|G\|_{\infty} := \sup_{x \in X} |G(x)| < \infty.$$

and the number $\|G\|_{\infty}$ is the least upper bound of $|G(x)|$ for all $x \in X$.



Definition 2.3

A sequence of complex-value functions $S_n : X \rightarrow \mathbb{C}$ converges uniformly to a complex-value function $S : X \rightarrow \mathbb{C}$ on a set X if

$$\lim_{n \rightarrow \infty} \|S_n - S\|_{\infty} = \lim_{n \rightarrow \infty} \sup_{x \in X} |S_n(x) - S(x)| = 0.$$



Exercise 2.1 If a sequence of complex-value functions $S_n : X \rightarrow \mathbb{C}$ converges uniformly to a complex-value function $S : X \rightarrow \mathbb{C}$ on a set X , then S is a bounded function on X .

Solution Since S_n converges uniformly to S on X , for any $\varepsilon > 0$, there exists an integer N such that for all $n \geq N$,

$$\|S_n - S\|_{\infty} = \sup_{x \in X} |S_n(x) - S(x)| < \varepsilon.$$

This implies that for all $x \in X$ and all $n \geq N$,

$$|S(x)| \leq |S_n(x)| + |S_n(x) - S(x)| < |S_n(x)| + \varepsilon.$$

Since S_N is a bounded function on X , there exists a constant M_N such that

$$|S_N(x)| \leq M_N \quad \text{for all } x \in X.$$

Therefore, for all $x \in X$,

$$|S(x)| < M_N + \varepsilon.$$

Thus,

$$\sup_{x \in X} |S(x)| \leq M_N + \varepsilon < \infty.$$

Hence, S is a bounded function on X .

 **Exercise 2.2** Show that if $F, G : X \rightarrow \mathbb{C}$ are bounded complex-value functions on a set X , then $\|F + G\|_\infty \leq \|F\|_\infty + \|G\|_\infty$ and $\|FG\|_\infty \leq \|F\|_\infty \|G\|_\infty$.

Solution For any $x \in X$, by the triangle inequality, we have

$$|F(x) + G(x)| \leq |F(x)| + |G(x)| \leq \|F\|_\infty + \|G\|_\infty.$$

Taking the supremum over all $x \in X$, we get

$$\|F + G\|_\infty = \sup_{x \in X} |F(x) + G(x)| \leq \|F\|_\infty + \|G\|_\infty.$$

Similarly, for the product, we have

$$|F(x)G(x)| = |F(x)| |G(x)| \leq \|F\|_\infty \|G\|_\infty.$$

Taking the supremum over all $x \in X$, we get

$$\|FG\|_\infty = \sup_{x \in X} |F(x)G(x)| \leq \|F\|_\infty \|G\|_\infty.$$

Definition 2.4 (Uniform Convergence of Series of Functions)

A series of complex-value bounded functions $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the sequence of partial sums

$$S_N(x) = \sum_{n=1}^N f_n(x)$$

converges uniformly on X . 

Definition 2.5 (Uniform Convergence of Products of Functions)

A product of complex-value bounded functions $\prod_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the sequence of partial products

$$P_N(x) = \prod_{n=1}^N f_n(x)$$

converges uniformly on X . 

Theorem 2.2 (Cauchy Criterion for Uniform Convergence of Series of Functions)

A series of complex-value bounded functions $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on a set X if the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges.

$$S_N(x) = \sum_{n=1}^N f_n(x)$$

satisfies

$$\|S_N - S\|_\infty \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$



Proof For any $\varepsilon > 0$, since the series of norms $\sum_{n=1}^{\infty} \|f_n\|_\infty$ converges, there exists an integer N such that for all

$m > n \geq N$,

$$\sum_{k=n+1}^m \|f_k\|_\infty < \varepsilon.$$

Then, for any $x \in X$ and all $m > n \geq N$, we have

$$|S_m(x) - S_n(x)| = \left| \sum_{k=n+1}^m f_k(x) \right| \leq \sum_{k=n+1}^m |f_k(x)| \leq \sum_{k=n+1}^m \|f_k\|_\infty < \varepsilon.$$

Taking the supremum over all $x \in X$, we get

$$\|S_m - S_n\|_\infty = \sup_{x \in X} |S_m(x) - S_n(x)| < \varepsilon.$$

This shows that the sequence of partial sums $\{S_N\}$ is uniformly Cauchy on X . Since the space of bounded functions on X is complete with respect to the supremum norm, it follows that $\{S_N\}$ converges uniformly to a bounded function $S : X \rightarrow \mathbb{C}$ on X . Hence, the series $\sum_{n=1}^\infty f_n(x)$ converges uniformly on X .

Theorem 2.3

A product of complex-value bounded functions $\prod_{n=1}^\infty (1 + f_n(x))$ converges uniformly on a set X if the series of norms $\sum_{n=1}^\infty \|f_n\|_\infty$ converges.

$$P_N(x) = \prod_{n=1}^N (1 + f_n(x))$$

satisfies

$$\|P_N - P\|_\infty \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Proof Let

$$P_N(x) = \prod_{k=1}^N (1 + f_k(x)), \quad x \in X.$$

Fix $\varepsilon > 0$. Since $\sum_{n=1}^\infty \|f_n\|_\infty$ converges, define

$$A := \sum_{n=1}^\infty \|f_n\|_\infty < \infty, \quad M := e^A.$$

For any $n \in \mathbb{N}$ and any $x \in X$, we have

$$|P_n(x)| = \prod_{k=1}^n |1 + f_k(x)| \leq \prod_{k=1}^n (1 + |f_k(x)|) \leq \prod_{k=1}^n (1 + \|f_k\|_\infty) \leq \exp\left(\sum_{k=1}^n \|f_k\|_\infty\right) \leq M.$$

Now choose $\delta > 0$ such that

$$e^\delta - 1 < \frac{\varepsilon}{M}.$$

Since $\sum_{n=1}^\infty \|f_n\|_\infty$ converges, there exists $N \in \mathbb{N}$ such that for all $m > n \geq N$,

$$\sum_{k=n+1}^m \|f_k\|_\infty < \delta.$$

For any $x \in X$ and $m > n \geq N$, we write

$$|P_m(x) - P_n(x)| = \left| \prod_{k=1}^n (1 + f_k(x)) \left(\prod_{k=n+1}^m (1 + f_k(x)) - 1 \right) \right|.$$

Using the inequality $|1 + z| \leq e^{|z|}$ for $z \in \mathbb{C}$, we obtain

$$\left| \prod_{k=n+1}^m (1 + f_k(x)) \right| \leq \exp\left(\sum_{k=n+1}^m |f_k(x)|\right),$$

and hence

$$\left| \prod_{k=n+1}^m (1 + f_k(x)) - 1 \right| \leq e^{\sum_{k=n+1}^m |f_k(x)|} - 1 < e^\delta - 1.$$

Combining the estimates, we obtain

$$|P_m(x) - P_n(x)| \leq M(e^\delta - 1) < \varepsilon.$$

Taking the supremum over all $x \in X$, we get

$$\|P_m - P_n\|_\infty = \sup_{x \in X} |P_m(x) - P_n(x)| < \varepsilon.$$

Thus $\{P_N\}$ is a uniformly Cauchy sequence on X . Since the space of bounded functions on X is complete with respect to the supremum norm, there exists a bounded function $P : X \rightarrow \mathbb{C}$ such that

$$\|P_N - P\|_\infty \rightarrow 0 \quad (N \rightarrow \infty).$$

Therefore, the infinite product $\prod_{n=1}^\infty (1 + f_n(x))$ converges uniformly on X .

Theorem 2.4 (Uniform Convergence and Zeros of Infinite Products)

Let $u_n : X \rightarrow \mathbb{C}$ be a sequence of bounded complex-value functions on a set X such that the series

$$\sum_{n=1}^\infty |u_n| \text{ converges uniformly on } X.$$

Then the infinite product

$$f(x) = \prod_{n=1}^\infty (1 + u_n(x))$$

converges uniformly on X .

Furthermore,

$$f(x_0) = 0 \text{ for some } x_0 \in X \iff u_n(x_0) = -1 \text{ for some } n \in \mathbb{N}.$$

i.e. f has a zero at x_0 if and only if one of the factors in the product is zero at x_0 .



Note We can use this theorem even in the case where u_n is constant function. And in this case, it states that

$$\sum_{n=1}^\infty |u_n| \text{ converges and } u_n \neq -1 \text{ for all } n \in \mathbb{N} \implies \prod_{n=1}^\infty (1 + u_n) \text{ converges to a nonzero limit.}$$

Exercise 2.3 Show that $e^x \geq x + 1$ for all $x \geq 0$

Solution By applying Mean Value Theorem to the function $f(x) = e^x$ on the interval $[0, x]$ for any $x \geq 0$, there exists some $c \in (0, x)$ such that

$$f'(c) = \frac{f(x) - f(0)}{x - 0} \implies e^c = \frac{e^x - 1}{x}.$$

Since $e^c \geq 1$ for all $c \geq 0$, we have

$$\frac{e^x - 1}{x} \geq 1 \implies e^x - 1 \geq x \implies e^x \geq x + 1.$$

Exercise 2.4 Show that for any sequence of complex numbers $\{a_n\}_{n=1}^\infty$, we have

$$\left| \prod_{n=1}^N (1 + a_n) - 1 \right| \leq \exp \left(\sum_{n=1}^N |a_n| \right) - 1.$$

Solution We prove this by induction on N . For $N = 1$, we have

$$|(1 + a_1) - 1| = |a_1| \leq e^{|a_1|} - 1, \text{ by Exercise 2.3.}$$

Now, assume the inequality holds for some $N \geq 1$. Then for $N + 1$, we have

$$\begin{aligned}
 \left| \prod_{n=1}^{N+1} (1 + a_n) - 1 \right| &= \left| (1 + a_{N+1}) \prod_{n=1}^N (1 + a_n) - 1 \right| = \left| (1 + a_{N+1}) \left(\prod_{n=1}^N (1 + a_n) - 1 \right) + a_{N+1} \right| \\
 &\leq |1 + a_{N+1}| \left| \prod_{n=1}^N (1 + a_n) - 1 \right| + |a_{N+1}| \\
 &\leq (1 + |a_{N+1}|) \left(e^{\sum_{n=1}^N |a_n|} - 1 \right) + |a_{N+1}|, \text{ by the induction hypothesis} \\
 &= e^{\sum_{n=1}^N |a_n|} - 1 + |a_{N+1}| e^{\sum_{n=1}^N |a_n|} - |a_{N+1}| + |a_{N+1}| \\
 &= (1 + |a_{N+1}|) e^{\sum_{n=1}^N |a_n|} - 1 \\
 &\leq e^{|a_{N+1}|} e^{\sum_{n=1}^N |a_n|} - 1, \text{ by Exercise 2.3} \\
 &= e^{\sum_{n=1}^{N+1} |a_n|} - 1.
 \end{aligned}$$

Thus, by induction, the inequality holds for all $N \in \mathbb{N}$.

Proof By our assumption, as $\sum_{n=1}^{\infty} |u_n|$ converges uniformly on X , the partial sums $S_N(x) = \sum_{n=1}^N |u_n(x)|$ converge uniformly to $S = \sum_{n=1}^{\infty} |u_n(x)|$ on X . Thus, for a given $\varepsilon > 0$, there is a N_0 (not depending on x) such that

$$\|S_N - S\|_{\infty} = \sup_{x \in X} \sum_{n=N+1}^{\infty} |u_n(x)| < \varepsilon, \quad \text{for all } N \geq N_0.$$

and exercise 2.1 implies that S is bounded on X . Hence, $S(x) = \sum_{n=1}^{\infty} |u_n(x)| \leq M$ for some $M > 0$ and all $x \in X$.

Let $P_N(x) = \prod_{n=1}^N (1 + u_n(x))$ denote the N -th partial product of $f(x)$. Then by exercise 2.4, we have

$$|P_N(x) - 1| \leq \exp \left(\sum_{n=1}^N |u_n(x)| \right) - 1 \leq e^M - 1 \quad \text{since } \sum_{n=1}^N |u_n(x)| \leq S(x) \leq M.$$

and thus $|P_N(x)| \leq e^M - 1 + 1 = e^M$ for all $x \in X$ and all $N \in \mathbb{N}$. Therefore, for any $M > N \geq N_0$ and any $x \in X$, using exercise 2.4 again, we have

$$|P_M(x) - P_N(x)| = |P_N(x)| \left| \prod_{n=N+1}^M (1 + u_n(x)) - 1 \right| \leq e^M \left(\exp \left(\sum_{n=N+1}^M |u_n(x)| \right) - 1 \right) < e^M (e^{\varepsilon} - 1) \quad (2.1)$$

for all $x \in X$. Taking the supremum over all $x \in X$, we get

$$\|P_M - P_N\|_{\infty} = \sup_{x \in X} |P_M(x) - P_N(x)| < e^M (e^{\varepsilon} - 1) < \varepsilon'$$

when we choose $\varepsilon = \log \left(\frac{\varepsilon'}{e^M} + 1 \right) > 0$. Thus, $\{P_N\}$ is a uniformly Cauchy sequence on X , and hence $f(x)$ converges uniformly on X .

Now, do the similar argument as in equation (2.1) and choose $M \rightarrow \infty$, $N = N_0$, we have

$$|f(x) - P_{N_0}(x)| \leq |P_{N_0}|(e^{\varepsilon} - 1) \quad \text{for all } x \in X.$$

Setting $\varepsilon < \frac{1}{2}$, we have

$$|f(x) - P_{N_0}(x)| < |P_{N_0}(x)| |\sqrt{e} - 1|$$

thus,

$$|f(x)| = |P_{N_0}(x) - (f(x) - P_{N_0}(x))| \geq |P_{N_0}(x)| - |P_{N_0}(x)| |\sqrt{e} - 1| = |P_{N_0}(x)| (2 - \sqrt{e}) \geq \frac{1}{3} |P_{N_0}(x)|$$

Therefore, if $f(x_0) = 0$ for some $x_0 \in X$, then $P_{N_0}(x_0) = 0$ and hence $u_n(x_0) = -1$ for some $1 \leq n \leq N_0$. The converse is trivial.

Theorem 2.5 (Weierstrass M-test for Infinite Series and Products)

If a sequence of complex function $u_n : X \rightarrow \mathbb{C}$ satisfies $|u_n(x)| \leq M_n$ and $\sum_{n=1}^{\infty} M_n$ converges, then the infinite series

$$\sum_{n=1}^{\infty} |u_n(x)| \text{ converges uniformly on } X.$$

hence the series

$$\sum_{n=1}^{\infty} u_n(x) \text{ converges uniformly on } X.$$

And by the previous theorem 2.4, the infinite product

$$f(x) = \prod_{n=1}^{\infty} (1 + u_n(x))$$

converges uniformly on X .



Proof Direct from the M -test in Complex Analysis and theorem 2.4.

Proposition 2.1 (Properties of Uniform Convergence)

1. uniform limit of continuous functions is continuous.
2. uniform limit of bounded functions is bounded. (See exercise 2.1.)
3. uniform limit of analytic functions is analytic.



Proof omitted.

Corollary 2.1 (Analyticity of Riemann Zeta Function and Euler Product)

Recall the Riemann zeta function and the Euler product formula

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad \text{and} \quad \mathcal{P}(z) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}} = \prod_{p \in \mathbb{P}} (1 + u_p(z)) \quad \text{for } \Re(z) > 1,$$

where $u_p(z) = \frac{1}{1-p^z} - 1 = \frac{p^{-z}}{1-p^{-z}}$.

Then both $\zeta(z)$ and $\mathcal{P}(z)$ are analytic on the half plane $\{z \in \mathbb{C} : \Re(z) > 1\}$. Further more, if

$$\zeta(z) = \mathcal{P}(z) \quad \text{for } \Re(z) > 1,$$

$|\zeta(z)| \neq 0$ on $\{z \in \mathbb{C} : \Re(z) > 1\}$ (i.e., $\zeta(z)$ has no zero on this half plane)



Note For a^z , where $a > 0$ is a positive real number, if not specified, we always consider it as $e^{z \log a}$, where $\log a$ is using the real logarithm to avoid multi-valued issue. And thus a^z is analytic on the whole complex plane $\mathbb{C}$.

Proof Notice that $\frac{1}{n^z}$ is an analytic function on $\mathbb{C}$ and $u_p(z)$ is analytic on $\{z \in \mathbb{C} : \Re(z) > 1\}$ for each prime p . Also,

$$|1 - p^{-z}| \geq 1 - |p^{-z}| = 1 - p^{-\Re(z)} \geq 1 - p^{-1} \geq \frac{1}{2} \quad \text{for } \Re(z) > 1.$$

Thus, for $\Re(z) \geq 1$, on the open set $X_\varepsilon = \{z \in \mathbb{C} : \Re(z) \geq 1 + \varepsilon\}$ for any $\varepsilon > 0$, we have

$$|u_p(z)| = \left| \frac{p^{-z}}{1 - p^{-z}} \right| \leq 2|p^{-z}| = 2p^{-\Re(z)} \leq 2p^{-(1+\varepsilon)} \quad \text{and} \quad \left| \frac{1}{n^z} \right| = n^{-\Re(z)} \leq n^{-(1+\varepsilon)}.$$

Since

$$\sum_{p \in \mathbb{P}} \frac{1}{p^{1+\varepsilon}} \leq \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} < \infty$$

the Weierstrass M-test 2.5 implies that both $\zeta(z)$ and $\mathcal{P}(z)$ converge uniformly on X_ε for any $\varepsilon > 0$. Therefore, by proposition 2.1, both $\zeta(z)$ and $\mathcal{P}(z)$ are analytic on $\{z \in \mathbb{C} : \Re(z) > 1\}$.

Further more, by theorem 2.4, since $u_p(z) \neq -1$ for all primes p and all z with $\Re(z) > 1$, we have $|\mathcal{P}(z)| \neq 0$ for all z with $\Re(z) > 1$. As $\zeta(z) = \mathcal{P}(z)$ for $\Re(z) > 1$, we have $|\zeta(z)| = |\mathcal{P}(z)| \neq 0$ for all z with $\Re(z) > 1$.

Note The same analysis can be applied to Dirichlet L -functions and their Euler product representations

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^z} \quad \text{and} \quad \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-z}}$$

but will discuss later once we have introduced Dirichlet characters.

2.2 Recap of Complex Analysis

Theorem 2.6 (Morera's Theorem)

Suppose $f : X \rightarrow \mathbb{C}$ is continuous function on an open set $X \subseteq \mathbb{C}$. If

$$\int_{\Gamma} f(z) dz = 0 \quad \text{for all loop } \Gamma \text{ in } X, \quad \text{then } f \text{ is analytic on } X.$$



Proof Omitted.

Theorem 2.7 (Cauchy Integral Theorem)

Suppose $f : X \rightarrow \mathbb{C}$ is an analytic function inside and on a simple closed contour Γ in an open set $X \subseteq \mathbb{C}$. Then

$$\int_{\Gamma} f(z) dz = 0$$



Proof Omitted.

Exercise 2.5 Use Morera's Theorem and Cauchy Integral Theorem to show the 3rd property in the above proposition 2.1.

Solution Suppose $\{f_n\}$ is a sequence of analytic functions on an open set $X \subseteq \mathbb{C}$ that converges uniformly to a function f on X . We want to show that f is analytic on X .

For any simple closed contour Γ in X , as f_n is analytic on X , by Cauchy Integral Theorem, we have

$$\int_{\Gamma} f_n(z) dz = 0 \quad \text{for all } n \in \mathbb{N}.$$

Since f_n converges uniformly to f on X , by the properties of uniform convergence, we can interchange the limit and the integral:

$$\int_{\Gamma} f(z) dz = \lim_{n \rightarrow \infty} \int_{\Gamma} f_n(z) dz = \lim_{n \rightarrow \infty} 0 = 0.$$

Thus, by Morera's Theorem, since f is continuous (as uniform limit of continuous functions is continuous) and the integral of f over any simple closed contour in X is zero, we conclude that f is analytic on X .

2.3 Euler Products

Definition 2.6

For a function $f : \mathbb{N} \rightarrow \mathbb{C}$ which is not identically zero, we say:

- (a) f is multiplicative, if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$.
- (b) f is completely multiplicative, if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$.



Exercise 2.6 Show that the Möbius function $\mu(n)$ which is defined as

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n = p_1 \cdots p_k \text{ is a product of } k \text{ distinct primes,} \\ 0 & \text{otherwise,} \end{cases}$$

is a multiplicative function.

Solution Let $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. We want to show that $\mu(mn) = \mu(m)\mu(n)$.

Case 1: If either m or n has a squared prime factor, then mn also has a squared prime factor. Thus, $\mu(m) = 0$ or $\mu(n) = 0$, and hence $\mu(mn) = 0 = \mu(m)\mu(n)$.

Case 2: If both m and n are products of distinct primes, say $m = p_1 p_2 \cdots p_k$ and $n = q_1 q_2 \cdots q_l$, where p_i and q_j are distinct primes. Since $\gcd(m, n) = 1$, the sets of primes $\{p_1, p_2, \dots, p_k\}$ and $\{q_1, q_2, \dots, q_l\}$ are disjoint. Therefore, $mn = p_1 p_2 \cdots p_k q_1 q_2 \cdots q_l$ is a product of $k + l$ distinct primes. Thus,

$$\mu(mn) = (-1)^{k+l} = (-1)^k (-1)^l = \mu(m)\mu(n).$$

Case 3: If either m or n is equal to 1, then without loss of generality, let $m = 1$. Then $\mu(1) = 1$, and we have

$$\mu(mn) = \mu(n) = \mu(1)\mu(n).$$

In all cases, we have shown that $\mu(mn) = \mu(m)\mu(n)$ for all $m, n \in \mathbb{N}$ with $\gcd(m, n) = 1$. Therefore, the Möbius function $\mu(n)$ is multiplicative.

Proposition 2.2

If $f : \mathbb{N} \rightarrow \mathbb{C}$ is a (completely) multiplicative function, and $f(n_0) \neq 0$ for some $n_0 \in \mathbb{N}$, then $f(1) = 1$.



 **Note** If f is identically zero, then $f(mn) = f(m)f(n)$ holds trivially for all $m, n \in \mathbb{N}$. Thus, if not specified, we always assume that a (completely) multiplicative function is not identically zero. And then by the above proposition holds always for any (completely) multiplicative function.

 **Exercise 2.7** Prove the above proposition.

Solution Since f is (completely) multiplicative, we have

$$f(n_0) = f(1 \cdot n_0) = f(1)f(n_0) \quad \text{since } \gcd(1, n_0) = 1.$$

Since $f(n_0) \neq 0$, we can divide both sides by $f(n_0)$ to get

$$f(1) = 1.$$

Lemma 2.1 (“Euler Product”)

Suppose $f : \mathbb{N} \rightarrow \mathbb{C}$ is a multiplicative function (not identically zero) and the series

$$\sum_{n=1}^{\infty} |f(n)| < \infty.$$

Then the “Euler Product” holds:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p \in \mathbb{P}} (1 + f(p) + f(p^2) + \cdots) = \prod_{p \in \mathbb{P}} \sum_{k=0}^{\infty} f(p^k).$$



 **Note** The last equality holds since f is multiplicative function (not identically zero) implies that $f(1) = 1$ by proposition 2.2.

Proof Let $u_p = \sum_{k \geq 1} f(p^k)$ for each prime p and set

$$P_N = \prod_{\substack{p \in \mathbb{P} \\ p \leq N}} (1 + u_p) = (1 + u_{p_1})(1 + u_{p_2}) \cdots (1 + u_{p_r}),$$

where $p_1, p_2, \dots, p_r$ are all the primes less than or equal to N and suppose $p_1 < p_2 < \cdots < p_r$. We need to show u_p is convergent for each prime p to ensure P_N is well-defined (i.e. The limit of P_N exists) for each N . As $|u_p| \leq \sum_{k=1}^{\infty} |f(p^k)| \leq \sum_{n=1}^{\infty} |f(n)| < \infty$, each u_p is convergent.

Notice that $f(1) = 1$ by proposition 2.2. Thus,

$$P_N = \lim_{K \rightarrow \infty} \left[\sum_{k_1=0}^K f(p_1^{k_1}) \sum_{k_2=0}^K f(p_2^{k_2}) \cdots \sum_{k_r=0}^K f(p_r^{k_r}) \right].$$

because if $\lim_{K \rightarrow \infty} a_K = A$ and $\lim_{K \rightarrow \infty} b_K = B$, then $\lim_{K \rightarrow \infty} a_K b_K = AB$. Rearranging the sums on the RHS, and using the multiplicative property of f , we have

$$\sum_{k_1=0}^K \sum_{k_2=0}^K \cdots \sum_{k_r=0}^K f(p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}) = \sum_{n \in F_{N,K}} f(n) \quad \text{since } \gcd(p_i^{k_i}, p_j^{k_j}) = 1 \text{ for } i \neq j.$$

where

$$F_{N,K} = \{n \in \mathbb{N} : n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}, 0 \leq k_i \leq K \text{ for } 1 \leq i \leq r\}.$$

Recall u_p is convergent, P_N exists for each N . Thus

$$\lim_{K \rightarrow \infty} \sum_{n \in F_{N,K}} f(n) = P_N.$$

Since $F_{N,K} \subseteq \mathbb{N}$, we can choose a large $M_{N,K}$ such that $F_{N,K} \subseteq \{1, 2, \dots, M_{N,K}\}$ for each N, K . Then

$$\left| \sum_{n=1}^{M_{N,K}} f(n) - \sum_{n \in F_{N,K}} f(n) \right| = \left| \sum_{\substack{n \leq M_{N,K} \\ n \notin F_{N,K}}} f(n) \right| \leq \sum_{n \geq N} |f(n)| + \sum_{n \geq 2^K} |f(n)| \quad (2.2)$$

where the last inequality holds since any $n \notin F_{N,K}$, it is either has a prime factor greater than N (So that $n > N$) or has a prime factor less than or equal to N with exponent greater than K (So that $n \geq p_i^K \geq 2^K$ for some $1 \leq i \leq r$).

Fix a value of N , and let $K \rightarrow \infty$, notice that

$$\lim_{K \rightarrow \infty} \sum_{n=1}^{M_{N,K}} f(n) = \sum_{n=1}^{\infty} f(n) \quad \text{and} \quad \lim_{K \rightarrow \infty} \sum_{n \in F_{N,K}} f(n) = P_N.$$

Thus, taking $K \rightarrow \infty$ in the above inequality (2.2), we obtain

$$\left| \sum_{n=1}^{\infty} f(n) - P_N \right| \leq \sum_{n \geq N} |f(n)|.$$

Finally, letting $N \rightarrow \infty$. As $\sum_{n=1}^{\infty} |f(n)|$ converges, the tail $\sum_{n \geq N} |f(n)| \rightarrow 0$ as $N \rightarrow \infty$. Thus,

$$\sum_{n=1}^{\infty} f(n) = \lim_{N \rightarrow \infty} P_N = \prod_{p \in \mathbb{P}} (1 + f(p) + f(p^2) + \dots).$$

Theorem 2.8 (Euler Product Formula)

Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be a completely multiplicative function (not identically zero) and suppose the series

$$\sum_{n=1}^{\infty} |f(n)| < \infty.$$

Then the Euler Product holds:

$$\sum_{n=1}^{\infty} f(n) = \prod_{p \in \mathbb{P}} (1 + f(p) + f(p^2) + \dots) = \prod_{p \in \mathbb{P}} \sum_{k=0}^{\infty} f(p)^k = \prod_{p \in \mathbb{P}} \frac{1}{1 - f(p)}.$$



Proof We only need to prove the last equality holds as the first two equalities are direct from lemma 2.1.

Recall the geometric series formula,

$$1 + z + z^2 + \dots = \frac{1}{1 - z} \quad \text{for } |z| < 1.$$

one may want to use this to justify the last equality. However, we need to ensure that $|f(p)| < 1$ for all primes p to apply the geometric series formula. But this is indeed true since if there exists a prime p_0 such that $|f(p_0)| \geq 1$, then as f is completely multiplicative, we have

$$|f(p_0^k)| = |f(p_0)|^k \geq 1 \quad \text{for all } k \in \mathbb{N},$$

thus the series $\sum_{n=1}^{\infty} |f(n)|$ diverges, as the subseries $\sum_{k=1}^{\infty} |f(p_0^k)|$ diverges. This contradicts our assumption that $\sum_{n=1}^{\infty} |f(n)| < \infty$. Therefore, we must have $|f(p)| < 1$ for all primes p , and hence we can apply the geometric series formula to get the last equality.

Corollary 2.2 (Classical Euler Product Formula)

The Riemann zeta function and the Euler product formula

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} \quad \text{and} \quad \mathcal{P}(z) = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}} \quad \text{for } \Re(z) > 1,$$

are equal, i.e.,

$$\zeta(z) = \mathcal{P}(z) \quad \text{for } \Re(z) > 1.$$



Proof By applying theorem 2.8 to the completely multiplicative function $f(n) = \frac{1}{n^z}$ for $\Re(z) > 1$, and noticing that

$\sum_{n=1}^{\infty} |f(n)| = \sum_{n=1}^{\infty} \frac{1}{n^{\Re(z)}}$ converges for $\Re(z) > 1$, we have

$$\sum_{n=1}^{\infty} \frac{1}{n^z} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{1}{p^z}} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-z}}.$$

Remark We will also want to apply $f(n) = \chi(n)n^{-z}$ to get the Euler product formula for Dirichlet L -functions later, where χ is a Dirichlet character.